Architecture of the Proposed AI-Driven Building Knowledge Intelligence System

Dual-Pipeline System Overview: Online Multi-Agent Hybrid RAG Runtime and Offline Knowledge Ingestion

PART I: ONLINE MULTI-AGENT HYBRID RAG INFERENCE PIPELINE (RUNTIME)

End-to-End Query Answering & Decision Support

1. User Query & Interaction
- Structural Engineer Query Input
- Domain: RCC Building Diagnostics
- React / Vite Client Application

HTTP Request

2. FastAPI Application Gateway
- Asynchronous REST API (/api/v1/agents/query)
- Input Normalization & Domain Guard
- Lifespan & Service State Management

Trigger Pipeline

LANGGRAPH MULTI-AGENT ORCHESTRATION CORE (5 SEQUENTIAL AGENTS)

Agent 1: Query Analysis
- Intent Classification
- Defect & Component NER
- Structural Scope Check
- Query Parameterization

Agent 2: Retrieval Agent
- Hybrid RAG Coordinator
- ChromaDB Vector Querying
- Neo4j Graph Querying
- Candidate Passage Pooling

Agent 3: KG Reasoning
- Multi-Hop Cypher Traversal
- Defect -> Cause -> Effect
- Entity Path Validation
- Structural Risk Inference

Agent 4: Recommendation
- Repair Action Synthesis
- Code-Compliant Workflows
- Urgency & Priority Rating
- Structural Safeguards

Agent 5: Report Synthesis
- Grounded Final Report
- Clause & Source Citations
- Evidence Synthesis
- Confidence Aggregation

Hybrid Retrieval Request

Grounded Inference Output

Grounded Output

Hybrid RAG & Context Fusion Engine
- HuggingFace all-MiniLM-L6-v2 (384-dimensional dense vectors)
- Dense Semantic Search over ChromaDB Collection
- Graph Pattern Matching over Neo4j Knowledge Graph
- Context Fusion: Vector Text Passages + Knowledge Graph Triples
- Calibrated Confidence Scoring (Deterministic Weighted Metric)

Fused Context

LLM Inference Subsystem (With Failover)
- Primary Provider: Groq LLM Inference Engine
  - Multi-API Key Rotation Pool (Accounts 1, 2, 3)
  - Automated HTTP 429 Rate Limit Rotation
- Fallback Provider: Google Gemini API (gemini-flash)
  - Automatic Failover when Groq quota is exhausted
- Citation & Evidence Grounding against Retrieved Sources

Vector & Graph Retrieval

PART II: OFFLINE KNOWLEDGE INGESTION PIPELINE & HETEROGENEOUS STORAGE

Verified Research Corpus, Document Processing, and Multi-Store Persistence

Authoritative Research Corpus
- 20 Validated Documents
- 355 Total Pages
- 5 Engineering Domains:
1. Concrete Testing (IS 516)
2. RCC Design (IS 456)
3. Seismic Detailing (IS 13920)
4. NDT Assessment (IS 13311)
5. Maintenance & Repair
- 105 Benchmark Evaluation Questions (20/20 Docs)
- Ground-Truth Citations

Raw PDFs

Document Processing & Chunking
- PyMuPDF Text & Structural Section Parsing
- Header, Table & Clause Boundary Recognition
- Recursive Character Splitting (1000 chars, 200 overlap)
- Source Document & Page Metadata Tagging

Parsed Chunks

Feature & Triple Extraction
- HuggingFace all-MiniLM-L6-v2 Embeddings (384-d)
- Dense Vector Generation for Semantic Search
- Structural Defect & Component Entity Extraction
- Knowledge Triple Extraction (Defect-Cause-Standard)

Vectors

Triples

Metadata

ChromaDB (Vector DB)
- Collection: infra_documents
- 1,066 Dense Vectors
- 384-d Cosine Metric Space
- HNSW Hierarchical Index
- Persistent Vector Index

Neo4j (Knowledge Graph)
- Neo4j Graph Database
- Structural Defect Entities
- Causal & Repair Relations
- Cypher Query Protocol
- Native Graph Traversal

MongoDB (Metadata Store)
- Database: infra_knowledge
- Document Master Records
- Processing Job Statuses
- User Profiles & RBAC
- Beanie Asynchronous ODM